

Cheng FANG

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Education Background

City University of Hong Kong, Hong Kong, China	09/2025-08/2029
Doctor of Philosophy in Data Science	
Core Modules: Data Mining and Knowledge Discovery, Social Foundations of Data Science, Advanced Statistics, Optimization	
Northwestern Polytechnical University, China	09/2022-03/2025
Master of Computer Science, GPA: 87.6/100	
Core Modules: Combinatorics, Mathematical Statistics, Data Mining Techniques, Internet of Things System Design	
Northwestern Polytechnical University, China	09/2018-07/2022
Bachelor of Data Science and Big Data Technology, GPA: 87.1/100	
Core Modules: Probability Theory and Mathematical Statistics, Computer Organization and Architecture, Operating System	

Publications

- [1] **C. Fang**, B. Guo, S. Liu, Z. Zhou, J. Tang, K. Ma, S. Luo, G. Song, Z. Yu. Responsive Test-Time Model Adaptation for Mobile Applications via Runtime-efficient Sparse Updates. *IEEE Transactions on Mobile Computing*, 2026, (CCF-A, Q1, IF: 7.7)
- [2] **C. Fang**, Z. Zhou, K. Ma, B. Guo. TaskIT: Memory-Efficient Fine-Tuning of Multi-LoRA LLMs via Cross-Task Importance Transfer. *CVPR 2026 (CCF-A)*.
- [3] **C. Fang**, S. Liu, Z. Zhou, B. Guo, J. Tang, K. Ma, Z. Yu. AdaShadow: Responsive Test-time Model Adaptation in Non-stationary Mobile Environments. *SenSys 2024 (CCF-B, Top-3 Conferences in Mobile Computing, Best Paper Honorable Mention)*.
- [4] **C. Fang**, B. Guo, S. Liu, K. Ma, Z. Yu. Weather-oriented Domain Generalization of Semantic Segmentation for Autonomous Driving. *UIC 2022 (CCF-C)*.
- [5] K. Ma, J. Tang, F. Dang, B. Guo, S. Liu, C. Fang, C. Fang, Z. Zhu, L. Wu, Y.-C. Chen, Z. Yu, Y. Liu. SURGEON: Memory-Adaptive Fully Test-Time Adaptation via Dynamic Sparsity Activation. *CVPR 2025 (Highlight paper, CCF-A)*.
- [6] S. Liu, B. Guo, **C. Fang**, Z. Wang, S. Luo, Z. Zhou, Z. Yu. Enabling Resource-efficient AIoT System with Cross-level Optimization: A survey. *IEEE Communications Surveys & Tutorials*, 2023 (Q1, IF: 35.6).
- [7] K. Ma, B. Guo, S. Liu, **C. Fang**, S. Luo, Z. Zheng, Z. Yu. AdaShift: Anti-collapse and Real-time Deep Model Evolution for Mobile Vision Applications. *IEEE Transactions on Mobile Computing*, 2024, (CCF-A, Q1, IF: 7.7)
- [8] J. Tang, H. Lu, R. Wu, X. Xu, K. Ma, **C. Fang**, B. Guo, J. Lu, Q. Chen, Y.-C. Chen. HAWK: Learning to Understand Open-World Video Anomalies. *NeurIPS 2024 (CCF-A)*.
- [9] S. Liu, W. Zhou, Z. Zhou, B. Guo, M. Wang, **C. Fang**, Z. Lin, Z. Yu. Deep Learning Inference on Heterogeneous Mobile Processors: Potentials and Pitfalls. *MobiSys 2024 Workshop*.
- [10] F. Qin, **C. Fang**, J. Li, W. Zhang. Object-specific Rapid Feature Adaption for One-shot Contour Primitive of Interest Extraction. *RCAR 2023*.

Research Experiences

Project Leader, Northwestern Polytechnical University, Xi'an, China	09/2022-10/2023
• Topic: Cross-level toolkit for efficient on-device DNN deployment. • Tasks: Developed hardware/software co-design algorithms to optimize resource utilization in DNN training and inference; Packed the algorithms into software libraries for diverse mobile and edge devices. • Output: Presented and demonstrated the toolkit at AIoTSys 2023 (extended version was accepted and published by MobiSys 2024 Workshop).	

Intern, Huawei, Xi'an and Shenzhen, China

07/2022-08/2023

- Topic: Edge-cloud collaborative domain mining and adaptation.
- Tasks: Designed an end-to-end hidden domain adaptation framework; Prototyped and evaluated the framework for real-time domain adaptation in robot campus inspection.
- Output: One paper accepted by IEEE Transactions on Mobile Computing.

Honors and Awards

Outstanding Graduate Student, Northwestern Polytechnical University	2024
First Class Scholarship (Postgraduate)	2023

Languages and Skills

Mandarin (native), English (fluent, IELTS: 6.5)), Python, C++